Personal Expense Tracker

Full-Stack Web Application Documentation & Setup Guide

PROJECT OVERVIEW

The Personal Expense Tracker is a responsive, full-stack application architected to help users record, categorize, and monitor their daily expenditures efficiently with instant data persistence in a PostgreSQL database.

CORE FEATURES

Expense Management

Record transactions instantly with specific descriptions, exact amounts, predefined categories, and dates.

Dynamic Tables & Totals

Real-time calculation of cumulative expenditures displayed cleanly in an interactive dashboard layout.

Database Persistence

Robust backend integration utilizing PostgreSQL for reliable storage, fast querying, and record deletion.

Responsive Design

Modern UI styled with Bootstrap framework ensuring full usability across desktop and mobile screens.

TECHNOLOGY STACK

- **Frontend:** HTML5, CSS3, Vanilla JavaScript, Bootstrap 5
- **Backend:** Node.js, Express.js, CORS middleware, dotenv
- **Database:** PostgreSQL (via `pg` node package)
- **Development Environment:** VS Code, Live Server, pgAdmin 4

PREREQUISITES & INSTALLATION GUIDE

Follow these structured steps to configure and run the application locally on your machine:

1. Database Setup (PostgreSQL)

1. Open **pgAdmin 4** and create a new database named `exptracker`.
2. Open the Query Tool for `exptracker` and execute the following SQL statement:

```
CREATE TABLE expenses (
    id SERIAL PRIMARY KEY,
    title VARCHAR(100) NOT NULL,
    amount NUMERIC(10, 2) NOT NULL,
    category VARCHAR(50) NOT NULL,
    date DATE NOT NULL
);
```

2. Backend Setup & Configuration

1. Open your terminal and navigate to the backend directory:

```
cd backend
```

2. Install required project dependencies:

```
npm install
```

3. Ensure your `.env` configuration file contains your exact database credentials:

```
PORT=3000
DATABASE_URL=postgresql://postgres:omar1234@localhost:5432/exptracker
```

4. Start the Express backend server:

```
node server.js
```

(Expected output: `Server running on port 3000`)

3. Frontend Execution

1. Open `frontend/Index.html` using **Live Server** in VS Code (typically running at `http://127.0.0.1:5500`).
2. Interact with the dashboard to seamlessly add and manage your expenses!